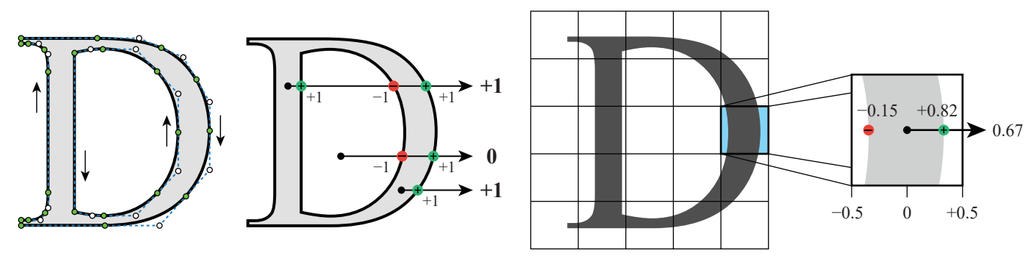
# The Slug Algorithm

sluglibrary.com

## Root Eligibility

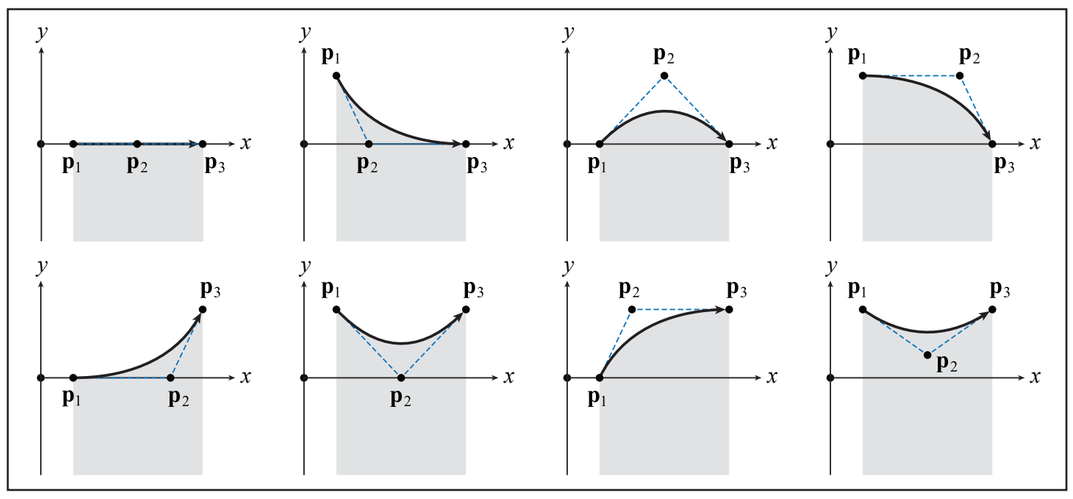
| Class | $y_3 < 0$ | $y_2 < 0$ | $y_1 < 0$ | Root 2 | Root 1 |
|---|---|---|---|---|---|
| A | 0 | 0 | 0 | 0 | 0 |
| B | 0 | 0 | 1 | 1 − | 0 |
| C | 0 | 1 | 0 | 1 − | 1 + |
| D | 0 | 1 | 1 | 1 − | 0 |
| E | 1 | 0 | 0 | 0 | 1 + |
| F | 1 | 0 | 1 | 1 − | 1 + |
| G | 1 | 1 | 0 | 0 | 1 + |
| H | 1 | 1 | 1 | 0 | 0 |
| | | | | 0x2E | 0x74 |

## Winding Number

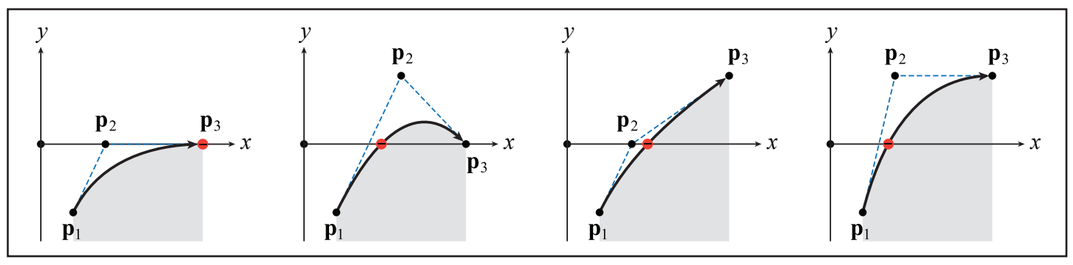



| | | |
|---|---|---|
| Quadratic Bézier curve | $\mathbf{p}(t) = (1-t)^2\mathbf{p}_1 + 2t(1-t)\mathbf{p}_2 + t^2\mathbf{p}_3$ $\quad \mathbf{p}_i = (x_i, y_i)$ | |
| Ray intersection equation | $p_y(t) = (y_1 - 2y_2 + y_3)t^2 - 2(y_1 - y_2)t + y_1 = 0$ | |
| Potential solutions | $t_1 = \frac{b - \sqrt{b^2 - ac}}{a}$ $\quad t_2 = \frac{b + \sqrt{b^2 - ac}}{a}$ $\quad \frac{d}{dt}p_y(t_1) \le 0$ $\quad \frac{d}{dt}p_y(t_2) \ge 0$<br>$a = y_1 - 2y_2 + y_3$ $\quad b = y_1 - y_2$ $\quad c = y_1$ | |
| Change to winding number for ray in positive $x$ direction | $+\text{sat}\left(kp_x(t_1) + \frac{1}{2}\right)$ if root 1 eligible +<br>$-\text{sat}\left(kp_x(t_2) + \frac{1}{2}\right)$ if root 2 eligible − | $k$ = pixels per em |
| Change to winding number for ray in negative $x$ direction | $-\text{sat}\left(\frac{1}{2} - kp_x(t_1)\right)$ if root 1 eligible −<br>$+\text{sat}\left(\frac{1}{2} - kp_x(t_2)\right)$ if root 2 eligible + | |

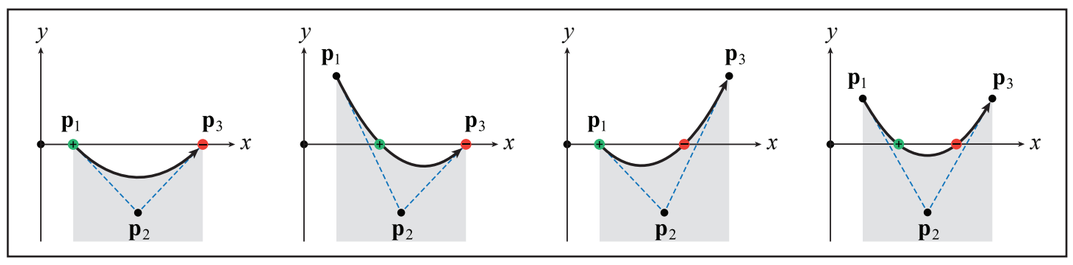
## Class A



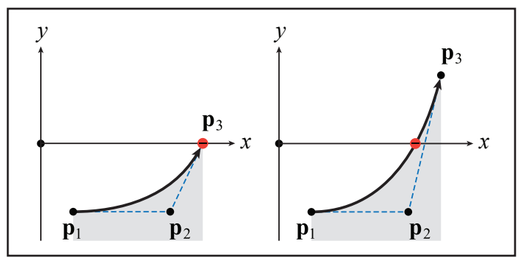

## Class B



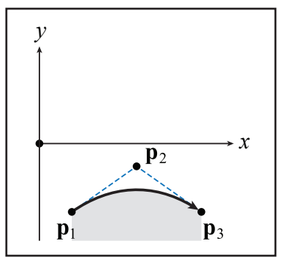

## Class C



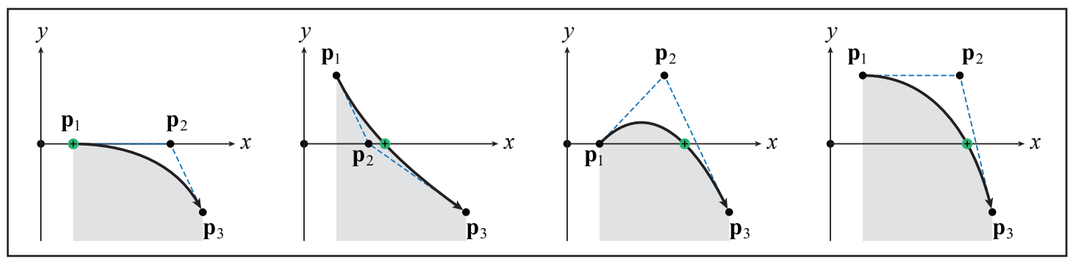

## Class D



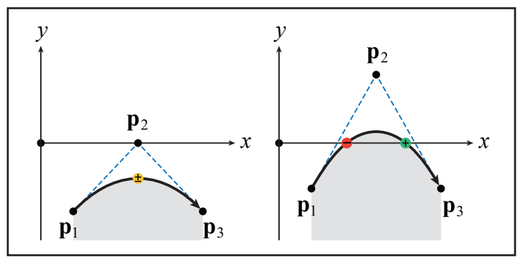

## Class H



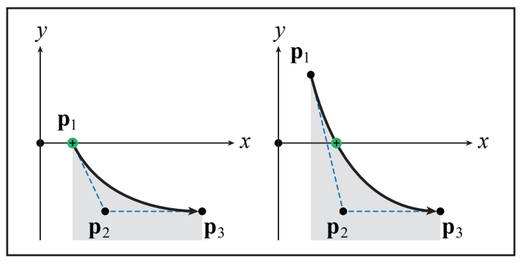

## Class E



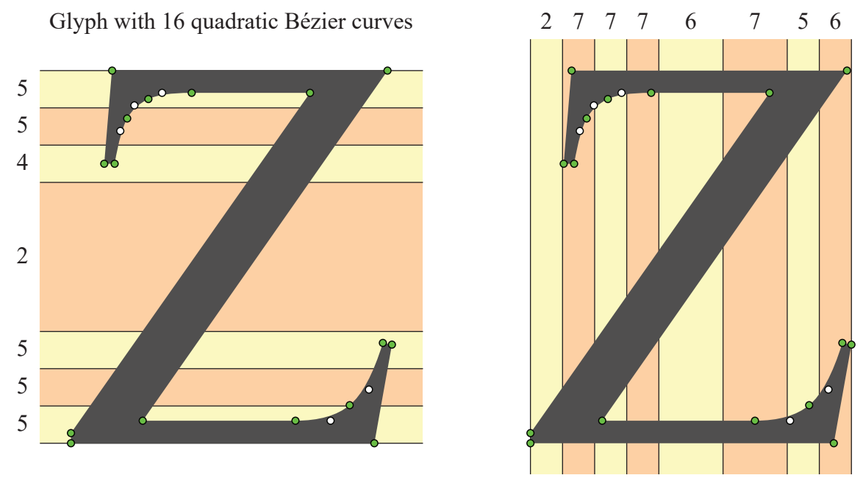

## Class F



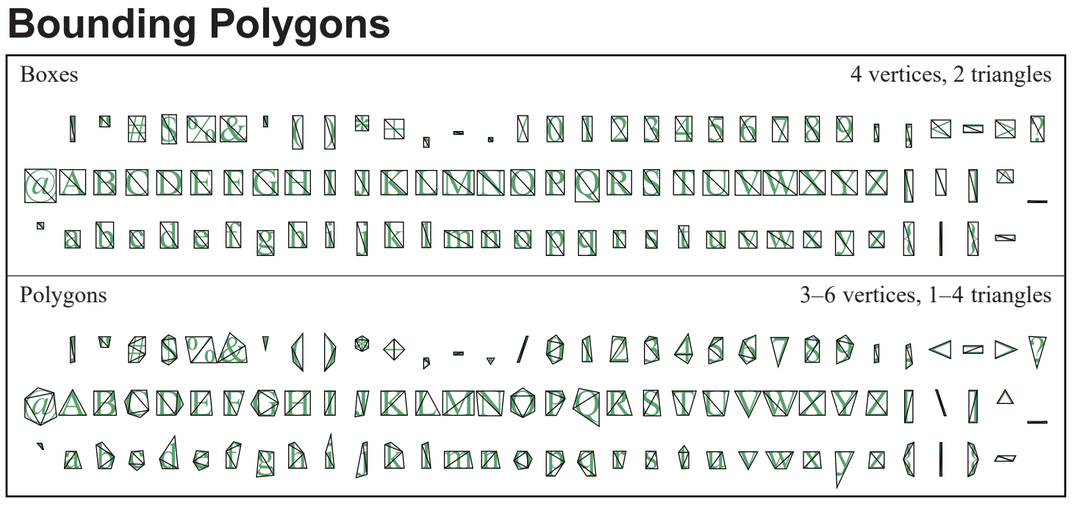

## Class G



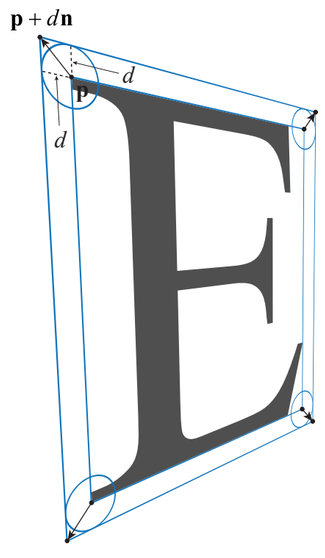

## Geometry Bands




## Bounding Polygons




## Dynamic Dilation




$\mathbf{m}$ = transformation matrix from object space to clip space

$w$ = viewport width $\quad h$ = viewport height

$d$ = object-space dilation distance

$(p_x, p_y)$ = object-space vertex position

$(n_x, n_y)$ = object-space vertex normal $\quad \hat{\mathbf{n}} = \mathbf{n}/\|\mathbf{n}\|$

$$\Delta x = \frac{w}{2}\left[\frac{m_{00}(p_x + d\hat{n}_x) + m_{01}(p_y + d\hat{n}_y) + m_{03}}{m_{30}(p_x + d\hat{n}_x) + m_{31}(p_y + d\hat{n}_y) + m_{33}} - \frac{m_{00}p_x + m_{01}p_y + m_{03}}{m_{30}p_x + m_{31}p_y + m_{33}}\right]$$

$$\Delta y = \frac{h}{2}\left[\frac{m_{10}(p_x + d\hat{n}_x) + m_{11}(p_y + d\hat{n}_y) + m_{13}}{m_{30}(p_x + d\hat{n}_x) + m_{31}(p_y + d\hat{n}_y) + m_{33}} - \frac{m_{10}p_x + m_{11}p_y + m_{13}}{m_{30}p_x + m_{31}p_y + m_{33}}\right]$$

$(\Delta x)^2 + (\Delta y)^2 = \frac{1}{4}$ (half-pixel viewport-space dilation)

$$s = m_{30}p_x + m_{31}p_y + m_{33} \qquad t = m_{30}\hat{n}_x + m_{31}\hat{n}_y$$

$$u = w\left[s(m_{00}\hat{n}_x + m_{01}\hat{n}_y) - t(m_{00}p_x + m_{01}p_y + m_{03})\right]$$

$$v = h\left[s(m_{10}\hat{n}_x + m_{11}\hat{n}_y) - t(m_{10}p_x + m_{11}p_y + m_{13})\right]$$

$$d = \frac{s^3 t + s^2\sqrt{u^2 + v^2}}{u^2 + v^2 - s^2 t^2}$$

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